

UNNAS HUSSAIN

Senior Software Engineer | Robotics | AI/ML | Control Systems

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TECHNICAL SKILLS

Programming Languages: Python, C++, C#, C, Java, Simulink, MATLAB, SQL/MySQL, Go Lang, Arduino, Racket, ACL2, LaTeX, JavaScript, CSS, HTML, Rust

Tools & Libraries: Git, ROS, Unix/Linux, Gazebo, CoppeliaSim, CMake, Windows Command Prompt, Unity, OpenCV, Isaac Sim, Docker, Gradle

Design: Autodesk, Solidworks, OnShape, KiCAD, Icarus Verilog, PSpice, BITalino

EXPERIENCE

Medtronic Inc., Advanced Surgical Technologies, *Boston, MA* Aug 2023 – Present
Senior Software Engineer - Controls

- Developing control algorithms, kinematic algorithms, and AI/ML computer vision software for endoluminal robots with continuum articulation joints. Delivering medical-grade accuracy at 4kHz response rates.
- Developing 6DOF Kalman Filter estimator for EM-enabled robotic bronchoscope navigating inside anatomy.
- Leading product development and architecture design for full software suite of robotic-assisted lung biopsies
- Implementing multi-threaded characterizing and tuning tools for real-time control loop tuning on QNX.
- Leading software efforts for ground-zero product development, building prototypes for formative studies across the world to engage surgeons and field experts on controls requirements, robot designs and usability.

Hologic Inc, *Marlborough, MA* Dec 2022 – July 2023
Embedded Software Engineer

- Implemented and updated embedded control logic for motorized and pneumatic subsystems within the ThinPrep 5000 (TP5000) system for fully automatic processing of gynecology, oncology, and Pap test samples.
- Developed slide designer to create custom laser-etched slides that are read by integrated Congex cameras.
- Designed robust TP5000 “power-on” tests to guarantee the system starts in a happy state or prompts user.

Amazon Robotics, *North Reading, MA* June 2021 – Nov 2022
Software Development Engineer

- Implemented software across navigation stack for over 520K Robotic Units, including task assignment with Min Cost Flow optimizers, AprilTag-based localization, and real-time driving adjustments for human interactions.
- Developed AWS Coral interceptors to compare computational load between base and optimized algorithms.
- Owned and developed internal tools to visualize live and historical metrics of robotic equipment, including real-time state updates for over 1500 daily users (Amazon Engineers and Warehouse Associates).

EDUCATION

Johns Hopkins University, *Baltimore, MD* Expected 2027
M.S.E in Robotics & Autonomous Systems, Part Time

Engineering for Professionals, Dynamics & Kinematics, General Robotics, Advanced Algorithms & Data Science

Northeastern University, *Boston, MA* May 2021
B.S.E in Computer Engineering & Computer Science with Honors GPA: 3.44

Robotics Science & Systems, Navigation, Dynamics, Computer Systems, Software Development, AI, OOD, Algorithms, Healthcare Technology, Electronics, Discrete Structures, Logic & Computation

SERVICE & VOLUNTEERING

FIRST Robotics Head Electrical Mentor, *Red Hawk Robotics (Melrose, MA) & The Nutrons (Boston, MA)*

- Mentored over 250 high school students to design, build, and code competition-level robot in 6-weeks.
- Taught student's various controls topics, such as ML pipelines, GPU-accelerated SLAM, and computer vision.

PATENTS, AWARDS, RECOGNITION

- SYSTEMS AND METHODS FOR CLINICAL WORKSPACE SIMULATION Patent # 20240256723
- HARDWARE-INDEPENDENT ADAPTIVE AUTO HOMING ALGORITHM Provisional Patent # 64/157,090
- SYSTEMS AND METHODS FOR FORCE SENSING AND ESTIMATION WITH CONTINUUM MANIPULATORS Provisional Patent # 64/122,849
- Medtronic Science & Technology Conference presenter 2024-2026
- Medtronic Technical Contributor of the Year Award nomination 2020
- Northeastern University Honors College Dean's List 2018 - 2021